TABULAR ML · FRAUD

"Deep learning is eating tabular data."

So I tested it on fraud detection — with confidence intervals.

swipe →

credit-bench

Bruno Karoubi

01 · THE SETUP

Tuned GBMs vs a modern transformer.

CONTENDERS

XGBoost, LightGBM & CatBoost **vs** FT-Transformer

DATA

IEEE-CIS Fraud · ~590K transactions · temporal holdout

a fair fight, not a demo

02 · THE RULE

Every comparison carries a confidence interval.

AUC DELTAS

DeLong 95% CIs (Sun-Xu, O(N log N))

PR-AUC DELTAS

block bootstrap

THE POINT

Significance — not one number on one split.

report the uncertainty, not just the score

03 · THE RESULT

Gradient boosting wins — decisively.

MODEL	HOLDOUT AUC
CatBoost	**0.914**
XGBoost	0.912
LightGBM	0.907
FT-Transformer	0.843

All three GBMs beat the default transformer by **6.4–7.1 AUC points** (p ≈ 0).

04 · THE NUANCE

Don't over-read a leaderboard.

CATBOOST VS XGBOOST

p = 0.39

Δ 0.0015 — a **statistical tie**. The top of the table is a coin flip.

---

a gap you can't measure isn't a gap

The discipline is the deliverable.

- Expanding-window temporal CV
- No early-stopping leakage into validation
- Pairwise CIs, calibration & SHAP disagreement

TAKEAWAY

- On credit/payments data, the deployed GBM stack still wins.
- Rigor — not vibes — is how you know it's real.
- "We benchmarked it" should mean confidence intervals.

Follow for more **measurement-first ML.**

credit-bench

Bruno Karoubi